

Synopsis

BEHAVIORAL SEGMENTATION FOR TARGETED MARKETING

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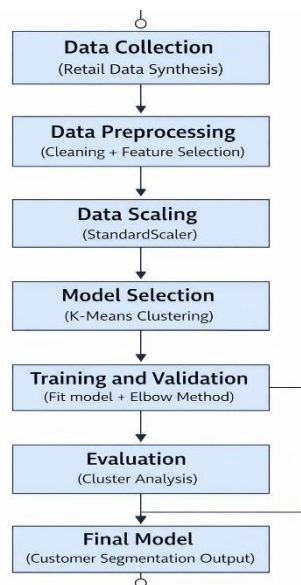
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Problem Description

Behavioral segmentation is a marketing strategy used to group customers based on their purchasing behavior and interaction patterns. By analyzing factors such as spending habits and purchase frequency, businesses can identify different types of customers and better understand their needs. This approach provides valuable insights that help organizations design personalized marketing strategies, improve customer engagement, and increase customer satisfaction. Companies can also use these insights to offer targeted promotions, loyalty programs, and product recommendations, which ultimately help in improving overall sales performance and building stronger customer relationships.

In this project, the objective is to segment customers based on their purchasing behavior using a machine learning approach. A synthetic dataset of 150 customers will be generated containing two key features: Annual Spending and Purchase Frequency. The dataset will be normalized using StandardScaler to bring the features to a similar scale. After preprocessing, the K-Means clustering algorithm will be applied to group customers into different clusters based on their behavior. The Elbow Method will be used to determine the optimal number of clusters. The resulting segments can help businesses understand customer behavior and develop more effective marketing strategies.

Steps of Implementation (ML Pipeline)



1. Data Collection

A synthetic dataset of 150 customers is generated using Python. The dataset includes features such as Annual Spending and Purchase Frequency, which represent customer purchasing behavior.

This data simulates real retail customer data that businesses collect to understand buying patterns.

2. Data Preprocessing

The collected data is prepared for analysis by standardizing the features using StandardScaler. This ensures that all variables have a similar scale and prevents one feature from dominating the clustering process.

Data preprocessing improves the performance and accuracy of machine learning algorithms.

3. Model Selection

The K-Means clustering algorithm is selected as it is an unsupervised learning method used to group similar data points into clusters based on their characteristics.

It is widely used in customer segmentation and pattern recognition problems.

4. Training and Validation

The K-Means model is trained on the scaled dataset. During training, the algorithm assigns each customer to the nearest cluster based on similarities in spending and purchase frequency.

This step allows the model to identify hidden patterns within the dataset.

5. Evaluation

The Elbow Method is applied to determine the optimal number of clusters by plotting inertia values against different cluster numbers.

The point where the curve bends (elbow point) indicates the best number of clusters for the dataset.

6. Final Model

After selecting the optimal cluster number, the final model segments customers into meaningful groups that can be used for targeted marketing strategies.

These segments help businesses design personalized offers and improve customer engagement.

Expected Modules / Libraries to be Used

NumPy

Used for numerical computations and for generating synthetic/random data for the dataset.

Pandas

Used to create, organize, and manage the dataset in a structured tabular format.

Scikit-learn

Provides machine learning tools such as StandardScaler for data normalization and K-Means clustering for customer segmentation.

Matplotlib

Used for data visualization such as plotting the **Elbow Method graph** and cluster scatter plots.

Seaborn

Used for advanced statistical data visualization and for creating more informative and visually appealing plots to analyze customer segmentation patterns.

Expected Evaluation Metrics and Plots

Scatter Plot Visualization

A scatter plot will be used to visually show the clusters formed by the K-Means algorithm. In this project, Annual Spending and Purchase Frequency will be plotted on the axes, where each point represents a customer. Different colors will represent different clusters. This helps in easily understanding how customers are grouped based on their purchasing behavior.

Elbow Method Plot

The Elbow Method is used to find the best value of K (number of clusters) in the K-Means algorithm. In this method, the model is run multiple times with different values of K, and the inertia value is calculated for each case. These values are then plotted on a graph. The point where the graph forms an “elbow” shape shows the optimal value of K, because after this point the improvement in clustering becomes very small.

Cluster Labels

After applying the clustering algorithm, each customer will be assigned a cluster label such as Cluster 0, Cluster 1, or Cluster 2. These labels represent the group to which the customer belongs. By analyzing these clusters, businesses can identify different types of customers based on their spending and purchasing patterns.

Required Topics From the Subject

Arrays

Arrays are used to store and manipulate dataset values efficiently during computations.

Functions

Functions help organize the code into smaller reusable parts, making the program easier to read and maintain.

Strings

Strings are used to handle textual data such as column names, labels, and other text-related operations in the dataset.

Regular Expressions (Regex)

Regex is used for pattern matching and processing text data, which can help in validating or cleaning dataset values if required.

Platform Required

Python

Python will be used as the main programming language for implementing the machine learning model.

Jupyter Notebook

Used to write and run the Python code step by step and visualize results easily.

VS Code

A code editor used for writing, editing, and managing the project files efficiently.

Kaggle

Kaggle can be used to explore datasets, run notebooks, and understand data science workflows.

GitHub

GitHub is used for version control and for storing and sharing the project code online.

Books and Link Sources

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
A comprehensive book for understanding machine learning concepts and practical implementation using Python.
2. Introduction to Machine Learning with Python – Andreas C. Müller & Sarah Guido
Provides a beginner-friendly introduction to machine learning techniques using Python and Scikit-learn.
3. Scikit-learn Official Documentation
<https://scikit-learn.org/stable/>
Used to understand the implementation of machine learning algorithms such as K-Means and preprocessing techniques.
4. Python Official Documentation
<https://docs.python.org/3/>
Used as a reference for Python syntax, libraries, and programming concepts.
5. Kaggle <https://www.kaggle.com/> used as a learning platform to explore datasets , notebooks , and practical data science projects.